Near in space, far in time: galaxy clusters seen through a turned spacetime

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Abstract

With the galaxy rule of Mathematical Materiality (MM), the radial acceleration relation with $a_0 = cH_0/6$, galaxy clusters lack mass. This record follows a premise of the author: the observer reads part of a cluster's spacetime as turned. Read as cos φ = r/R_b, the turn gives an effective scale that grows towards the centre as A = R_b/r. Its exponent is 1 between centre and edge, zero in galaxies, and the same for light and for matter. The main tests were registered before their computations, checks added after an outcome are marked as such, and every step was checked by two AI assistants. What the data say: What does not hold, or is not decidable: What is not claimed. The turn is a coherent reading of these data, not a proof. An ordinary explanation (NFW-like mass with a baryon fraction rising outwards) gives the same mean exponent, and what fixes R_b is open. The four registered tests of MM 2.46 for future data are unchanged.

Near in space, far in time: galaxy clusters and a turned spacetime

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The archived text is in English.

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How to read this text. First we explain the words that are needed. Then comes a summary, *In short*. Sections 1 to 6 go into the details. At the end you find how the work was done, a glossary and the references. Numbers like 2.46 or 2.70 refer to other works of the author, archived on Zenodo.

Before we start: the words we need

Galaxies, clusters and groups. A *galaxy* is a huge collection of stars, gas and dust held together by gravity. The Milky Way is our galaxy. *Disc* galaxies are flat and rotate around their centre. They do not rotate all in one piece like a merry-go-round: each star follows its own orbit. A *galaxy cluster* is a very large group of galaxies bound by gravity. Between the galaxies of a cluster there is a very hot gas, which shines in X-rays. A *galaxy group* is like a cluster, but smaller.

Acceleration. Acceleration says how much the speed changes every second. A star that goes round the centre of a galaxy has an acceleration towards the centre. Gravity keeps it in orbit.

The missing mass. From the motion of stars and gas one obtains how much mass there is. This is the *total mass*. The matter we see, stars and gas, is called *baryonic matter*: it is made of atoms. In galaxies and clusters the total mass is much larger than the baryonic mass. There are two kinds of answer:

- there is an invisible matter, *dark matter*;
- or gravity works differently when the acceleration is very small.

MOND and the scale a_0. MOND is the second answer. It is an idea of the physicist Milgrom. There is a special, tiny acceleration, called a_0 (read *a-zero*). Above a_0 Newton's gravity holds. Below a_0 gravity is stronger than Newton's.

The galaxy rule. In disc galaxies an observed law holds: the *radial acceleration relation*, RAR for short. It links two accelerations:

- g_obs, the observed acceleration, obtained from the motion;
- g_bar, the acceleration that baryonic matter alone would give.

The formula is g_obs = g_bar/(1 − e^(−√(g_bar/a_0))). In words: when g_bar is much larger than a_0, g_obs is almost equal to g_bar, as Newton requires. When g_bar is smaller than a_0, g_obs is larger. In MM the scale is $a_0 = cH_0/6$. Here c is the speed of light and H_0 is the *Hubble constant*, which says how fast the universe is expanding today.

The effective scale. In clusters, even with the galaxy rule, mass is missing. One can pretend that the scale a_0 is larger. The enlarged scale is written $A \cdot a_0$. The number A says how many times a_0 has to be multiplied for the numbers to add up. If A = 1, the galaxy rule is enough.

The centre and the edge. In a cluster A is measured at two points: near the centre and towards the edge. They are called A_c (centre) and A_d (edge). Distances from the centre are given in units of a reference radius, R500. It is the radius within which the mean density of the cluster is 500 times the critical density of the universe. The *critical density* is a reference density, linked to the speed at which the universe expands. In the same way r2500 is a more inner radius, with 2500 times, and R200m is a more outer radius, with 200 times the mean density of the universe.

The exponent. Often a quantity changes with distance as a power. If A falls as 1/r, the exponent is said to be 1. An example with numbers chosen by us: with exponent 1, if the radius doubles, A halves. With exponent 0, A does not change. Between centre and edge the exponent is computed from the two values of A and the two radii.

Median. It is the value that sits in the middle: half of the cases are below, half above.

Gravitational lensing. Mass bends the light that passes close to it, like a lens. So a cluster can be weighed by looking at how it deforms the images of the galaxies behind it. Lensing sees all the mass, even the invisible part.

Hydrostatic gas. The hot gas of a cluster pushes outwards with its pressure. Gravity pulls it inwards. When the two pushes balance, the gas is said to be *in hydrostatic equilibrium*. From the temperature and the density of the gas the mass is then obtained: it is the *hydrostatic mass*. Usually this mass comes out a little lower than the true one. This is the *hydrostatic bias*.

The words of the MM framework. Mathematical Materiality, MM for short, is the author's framework of ideas.

- A *premise* is a starting sentence of the author: a fixed point, like a postulate.
- A *reading* is an interpretation, written so that it can be put to the test. One also says that a body *reads* something when it takes it into account.
- The *link* is the starting relation of MM: the relation between two distinct terms.
- The *matrix* is reality counted from one point, that is, as an observer counts it.
- The *distal end* is the far end of a link.
- *Registered* means written and fixed with its date before doing the computations. So it cannot be changed afterwards.

In short

The problem. The galaxy rule of MM is used, with $a_0 = cH_0/6$. Galaxy clusters lack mass. The usual way to say this is: in clusters a single effective scale is needed, about ten times a_0.

The idea of this record. The record follows instead a premise of the author. The observer reads part of a cluster's spacetime as **turned**. *Spacetime* is space and time taken together, as in relativity. With this turn the scale grows towards the centre.

An everyday example. You look at a pencil held crosswise, turned a little towards you. It looks shorter than it is. Part of its length goes towards you, and you do not see it. The premise says something similar for the space and time of a cluster.

The formula. The turn is read as cos φ = r/R_b.

- φ (read *phi*) is the angle of the turn. The *cosine*, cos φ, says how much of a turned arrow remains in the starting

direction. For a zero angle it is 1. For a right angle it is 0.
- r is the distance from the centre of the cluster. R_b is the distal end of the link.
- This gives an effective scale $A \cdot a_0$ with **A = R_b/r**.

What it means:

- between centre and edge the exponent is 1;
- in galaxies the exponent is zero;
- the same holds for light and for matter.

The main tests were registered before their computations. The checks added after an outcome are marked as such. Every step was checked by two AI assistants.

The results, in six points.

1. **The exponent is close to 1 in the median.** The sample is 19 clusters: 12 from the X-COP sample and 7 from Vikhlinin et al. 2006. *Et al.* means *and others*. The exponent of A between centre and edge ranges from 0.62 to 1.62. The median is 1.02. At the centre A is 2.2 to 9.4 times larger than at the edge. In disc galaxies, instead, the exponent is nearly zero: from 0.05 to 0.33 in MM 2.46.1 (premise 48). The value depends on the stellar *mass-to-light ratio*, that is, on how much mass there is for each unit of light.

1. **Lensing and gas are compatible with the same reading.** The sample is 16 CLASH clusters, measured at the radius r2500.
2. With gravitational lensing A ≈ 12.5. The sign ≈ means *about equal*.
3. With hydrostatic gas A ≈ 8.9.
4. The ratio between the two masses is 0.87: it is the usual hydrostatic bias.
5. There was also a variant in which only slow matter feels the turn. With lensing it would give A ≈ 1. This variant fails.
6. Careful: this is a registered re-reading of values already computed. And dark matter predicts the same result.

1. **Not exactly 1 in every cluster.** With the statistical errors of X-COP, the reading "exponent 1 in each cluster" fails. χ^2 measures how far a model is from the data: here it is 174 for 12 degrees of freedom, much more than expected if the model is good (section 3.3). The clusters scatter by about ±0.2 around 0.96. But there is a limit. If the centre-to-edge hydrostatic *systematics* change from cluster to cluster by 10-15 %, the test becomes not decidable. Systematics are errors that always shift the same way, like a badly calibrated scale.

1. **What sets the exponent.** Mostly the slope of the total mass between centre and edge. The Spearman coefficient, a number between −1 and +1 that says how much two quantities go together, is −0.875 over 19 clusters. In part this happens by construction, that is, because of how the computations are made. The slope of the baryonic mass passes the threshold over 19 clusters (−0.504). But it passes it only because it goes together with the slope of the total mass. Within X-COP it does not pass it (−0.245). There is no robust association with the stars or with the redshift, which says how far back in time we see the cluster.

1. **Where the missing part sits.** Take the mass that is missing under the galaxy rule. In 6 of the 7 X-COP clusters with stellar profiles, this mass follows the stars of the member galaxies, not the gas. Groups, at the same radius, lack slightly less mass than clusters. There are 16 groups; the missing factor F is about 1.8 (F ≈ 1.8) against 1.95.

1. **What does not hold, or is not decidable.**
2. The edge R_b = R200m: the amplitude is not decidable, and the system-by-system prediction is not supported.
3. The angle of view of the cluster's long axis, at r2500: not decidable.
4. A rule that obtains the *depth* from the gas temperature: not decidable.
5. The cluster as a full external context: this reading fails.
6. One reading is compatible, but it was found after seeing the data. It is an "epoch-equivalent" a_0: for the median cluster it corresponds to z ≈ 4.3 (if $a_0 \propto H$) or to 3.2 (if $a_0 \propto 1/t$). We look at it in section 3.8.

What all this means. The turn is a coherent reading of these data. It is not a proof. An ordinary explanation gives the same mean exponent: an NFW-like total mass (section 1), with a baryon fraction that rises outwards. And what fixes R_b remains open.

1. What a physicist will recognise

This section lists the things already known, from which the record starts.

- **The galaxy rule.** It is the formula g_obs = g_bar/(1 − e^(−√(g_bar/a_0))) of McGaugh, Lelli and Schombert (2016). Here $a_0 = cH_0/6 = 1.0914 \cdot 10^{-10}$ m s^{-2} is used, with $H_0 = 67.4$ (MM 2.46).
- m s^{-2} means metres per second squared, the unit of accelerations.
- 10^{-10} means divided by ten billion.
- H_0 is measured in kilometres per second per megaparsec.
- **The mass deficit in clusters.** MOND-like laws, in clusters, give too little mass. A factor of about two is missing, concentrated at the centre. This was shown by Sanders (1999, 2003), Pointecouteau and Silk (2005), Famaey and McGaugh (2012), Kelleher and Lelli (2024).
- **One scale for clusters.** One scale can be imposed on all clusters. Then a cluster "RAR" is obtained, with a scale about ten times higher. Tian et al. (2020) find $2.0 \cdot 10^{-9}$ m s^{-2}.
- **The effective scale falls with radius.** Inside clusters, A becomes smaller going towards the edge. MM 2.46 found a median slope of −0.97 in 12 X-COP clusters.
- An ordinary explanation gives the same result. In the *deep regime*, that is, at accelerations far below a_0, d ln A/d ln r ≈ 2m − b − 2 holds.
- Here m is the slope of the total mass and b that of the baryonic mass. The slope says by how much the mass grows when the radius grows.
- For NFW-like masses, with a baryon fraction that rises outwards, the result is about −1. NFW is a mass profile typical of dark-matter halos in computer simulations.
- **Hydrostatic masses are low.** They are 10-40 % lower than the true value (Kelleher and Lelli 2024 and the works they cite). In CLASH clusters the lensing masses and the X-ray masses agree within this range (Donahue et al. 2014; Umetsu et al. 2016).

2. The reading: a turned spacetime

The author's premises

Here are the author's premises from which the record starts. The number says which premise it is. The sixth, the third and the fifth are given in the author's exact words; the seventh is summarised.

- **The sixth:** «prova a pensare una rotazione del nesso internamente alla matrice come se certi nessi il descrittore avesse un rapporto frazionale con il descritto che equivarrebbe ad una sostanziale rotazione dello spazio tempo percepito dall'osservatore, non prendere alla lettera le mie parole ma estrai il senso» ("try to think of a rotation of the link inside the matrix, as if for some links the describer had a fractional ratio to the described, which would amount to a substantial rotation of the spacetime perceived by the observer; do not take my words literally, extract the sense").
- The *describer* is the one who describes, the observer. The *described* is what is described.
- A *fractional ratio* is a ratio that is a fraction, like a half or a third.
- **The third:** «la galassia riguarda lo spazio l'ammasso relativamente più il tempo» ("the galaxy concerns space, the cluster relatively more time"). And then: «l'ammasso è un tempo lontano che nella prospettiva dell'osservatore sembra vicino. nello spazio» ("the cluster is a far time that, in the observer's perspective, looks near in space").
- **The fifth, in its follow-up:** «da valutare non solo per gli oggetti fisici ma anche per lo spazio tempo stesso» ("to be assessed not only for physical objects but for spacetime itself").
- **The seventh, on method.** Start from the centre-to-edge ratio of each cluster. Look at how much it varies at most and at least. Know that in single galaxies it is almost zero. Do not try to rebuild the observational matrix.

The translation into a formula

The translation of the premises into a formula is the AI's. It was registered before the computations.

- **The fraction** is cos φ = r/R_b. Here R_b is the distal end of the link. It is not the cluster's outskirts.
- **The effective scale** is a_0/cos φ. So A = R_b/r.
- **In clusters.** If R_b is fixed in a cluster, the centre-to-edge exponent of A is exactly 1.
- **In galaxies.** In a galaxy the distal end reaches the *cosmic horizon*, that is, the limit of the universe that we can see (MM 2.46). Then A = 1 and the exponent is 0.

An example with numbers chosen by us. The centre is at 0.2 R500 and the edge at 1.0 R500. With exponent 1, A at the centre is 5 times A at the edge: 1.0 divided by 0.2 makes 5.

The same law, read in another way. It is the same law as the "fixed distal end" of MM 2.46, a = A/r. There it was read as a fixed end. Here it is read as a turn of the perceived spacetime. The new content of this record is in the tests, not in the formula.

How A is measured

- At each radius, A solves the equation ν(y/A)·M_bar = M_tot.
- ν(y) = 1/(1 − e^(−√y)) is the *boost* of the galaxy rule: how many times gravity is stronger than Newton's. The letter ν is read *nu*.
- y = g_bar/a_0.
- M_bar is the baryonic mass and M_tot the total mass.
- In words: A is the number by which a_0 has to be enlarged so that the baryonic mass, with the boost, gives the total mass.
- F = M_tot/(ν(y)·M_bar) is the *missing factor* at fixed a_0. It says how many times the total mass exceeds the one given by the galaxy rule.
- The exponent between two radii is s = ln(A_c/A_d)/ln(r_d/r_c). It is the mean exponent between the two points, not a slope at a single point. ln is the *natural logarithm*.

3. Results

3.1 How much A changes between centre and edge

For X-COP the masses, the gas and the stars come from Eckert et al. 2019, Ettori et al. 2019, Ghirardini et al. 2019 and Ghizzardi et al. 2021. X-COP is a sample of clusters studied in depth with X-ray satellites.

sample	radii (R500)	A_c/A_d, range (median)	exponent of A, range (median)	F_c/F_d, median
X-COP, 12	0.2 → 1.0	2.7-9.4 (4.5)	0.62-1.39 (0.93)	1.92
X-COP, non-thermal pressure	0.2 → 1.0	2.4-8.3 (4.0)	0.55-1.31 (0.86)	1.80
Vikhlinin et al. 2006, 7	0.40-0.48 → 1.0	2.2-3.5 (3.0)	0.85-1.62 (1.47)	1.63
all 19		2.2-9.4 (3.7)	0.62-1.62 (1.02)	1.79

Let us read the table row by row.

- **X-COP, 12 clusters**, between 0.2 and 1.0 R500. The ratio A_c/A_d ranges from 2.7 to 9.4, with median 4.5. The exponent ranges from 0.62 to 1.39, with median 0.93. F_c/F_d has median 1.92.
- **X-COP with non-thermal pressure.** Besides heat, the motions of the gas also hold it up: this is the *non-thermal pressure*. Taking it into account, A_c/A_d ranges from 2.4 to 8.3 (median 4.0). The exponent ranges from 0.55 to 1.31 (median 0.86). F_c/F_d has median 1.80.
- **Vikhlinin et al. 2006, 7 clusters**, between 0.40-0.48 and 1.0 R500. A_c/A_d ranges from 2.2 to 3.5 (median 3.0). The exponent ranges from 0.85 to 1.62 (median 1.47). F_c/F_d has median 1.63.
- **All 19.** A_c/A_d ranges from 2.2 to 9.4 (median 3.7). The exponent ranges from 0.62 to 1.62 (median 1.02). F_c/F_d has median 1.79.

What one gets from it.

- The galaxy and cluster ranges are separated.
- Here there are no systems in between measured at two radii.
- Whether the change is a jump or continuous is not established.
- The X-COP part repeats, with another *estimator* (another way of computing), the slope already published in MM 2.46.

3.2 What sets the exponent

An approximate formula. In the deep regime A ∝ M_tot²/(r²·M_bar). The sign ∝ means *proportional to*. So s ≈ 2 − 2m + b, where m and b are the mean slopes of the total mass and of the baryonic mass.

- This formula describes the measured exponent well.
- The Spearman coefficient between formula and measurement is 0.991. The largest difference is 0.077.
- The *Spearman coefficient* is a number between −1 and +1. It says how much two quantities grow together, in the same order. Near +1 they grow together; near −1 one grows when the other falls.

The total mass weighs more. Most of the differences between clusters come from the slope of the total mass.

- Spearman(s, m) is −0.875 over 19 clusters and −0.825 over the X-COP clusters.
- Over the 7 Vikhlinin clusters it is −0.714: below the threshold, which is 0.786.
- This dominance is largely due to the factor 2 in the formula: m and b have similar variances. The *variance* measures how spread out the values are.
- More than half of the variance of 2m over 19 clusters comes from the difference between the two samples. The two samples are measured over different radial ranges and are selected differently.

The baryonic mass.

- Spearman(s, b) = −0.504 passes the threshold (0.456) over 19 clusters.
- But b varies together with m: their correlation is 0.85. And b enters s with the opposite sign.
- Within X-COP it is −0.245, below the threshold.

No other robust association. There is no robust *monotonic* association, that is, always in the same direction, with:

- the stellar slope (7 clusters);
- the stellar fraction, which in 12 of the 19 clusters is fixed by construction;
- the redshift. The *redshift*, z, says how much the light has been stretched by the expansion of the universe: the larger it is, the farther back in time the object is.

A caution. A shallow slope of the total mass can be read as "concentration". And concentration can be read as an early formation epoch. But this is an interpretation. The link between concentration and epoch comes from standard simulations, not from MM.

3.3 Is the exponent 1 in every cluster?

Registered outcome: fails.

- s = 1 is put to the test in each cluster. $\chi^2(s = 1)$ is 174 for 12 degrees of freedom.
- χ^2 (chi-squared) measures how far a model is from the data. The *degrees of freedom* count the independent comparisons. If the model is good, χ^2 is close to the number of degrees of freedom. Here 174 is much more than 12.
- The statistical errors of the X-COP files were used, with 1000 *log-normal* draws: random numbers with the error shape suited to quantities that are always positive.
- The error σ (sigma) on the exponent ranges from 0.03 to 0.11.
- The *intrinsic* scatter, that is, the true one between the clusters, is 0.21 around μ = 0.96. It is the most probable value, called the *maximum-likelihood* value. μ (read *mu*) is the mean.
- With non-thermal pressure χ^2 is 223.

This outcome hits the reading "exponent 1 in each cluster", not the MM framework.

A check added after the outcome, declared as such.

- The X-COP errors are statistical only: from 1 to 7 % (1-7 %) on M_tot.
- Suppose that the centre-to-edge ratio of the hydrostatic mass changes from cluster to cluster by 10 %. Then χ^2 reaches the limit of the "fails" zone.
- With 13 % it reaches the limit of the "agreement" zone.
- With non-thermal pressure 11 % and 15 % are needed.
- This is not the absolute hydrostatic bias. It is a change along the radius that differs from cluster to cluster. It is not measured here.

The reading of the result.

- The geometry of the turn fixes the sample mean of the exponent at 1. It does not fix each cluster's profile.
- The reading "R_b fixed and cos φ = r/R_b in each cluster" fails in the tested form.
- One could give each cluster its own exponent: cos φ = (r/R_b)^s. But it is an untested extension. It adds one parameter per cluster. Until something fixes s, it describes the profiles rather than predicting them.

3.4 Light and gas

The data. They are 16 CLASH clusters, at the radius r2500. CLASH is a large programme of observations of clusters. The images for lensing come from the Hubble Space Telescope; the hot gas is observed with X-ray telescopes.

- The lensing masses come from the NFW fits of Umetsu et al. 2016.
- The hydrostatic masses and the gas fractions come from Donahue et al. 2014.
- The stars are estimated from the median X-COP ratio between stars and gas.

An important limit. The A values were already computed when the zones of the test were written. So this is a registered re-reading, not a blind test.

quantity	value	registered zone
A from lensing, median	12.5 (above 2 in 16 of 16)	"only slow matter reads the turn" (A ≈ 1 from lensing): **fails**
M_X/M_L, median	0.87 (0.64-1.29; 16 of 16 in 0.5-1.3)	"spacetime is turned" (light and gas read the same): **agreement**
A from gas, median	8.9 (ratio of medians 0.72: A grows roughly as the mass squared)	

Let us read the table.

- **A from lensing:** median 12.5. It is above 2 in 16 clusters of 16. The variant "only slow matter reads the turn" would give A ≈ 1. This variant fails.
- **M_X/M_L** is the ratio between the X-ray mass (gas) and the lensing mass (light). The median is 0.87. The values range from 0.64 to 1.29, and 16 of 16 lie between 0.5 and 1.3. The variant "spacetime is turned", in which light and gas read the same, is in **agreement**.
- **A from gas:** median 8.9. The ratio of the medians is 0.72. In words: A grows roughly as the square of the mass.

What it means.

- If the turn is the right reading, it concerns the spacetime that is read, as the fifth premise says. It does not concern only the motion of matter.
- But lensing sees the missing mass also if there is dark matter. So the test T-R1 separates the two variants of the turn. It does not separate the turn from dark matter.

3.5 Where is the distal end?

The rule tested. R_b = R200m, the radius enclosing 200 times the mean density of the universe. In this rule no number is fitted to the sample, within the NFW fits and the adopted cosmology (Ω_m = 0.3). Ω_m is the fraction of matter in the universe.

There are 50 systems: 12 X-COP, 6 Vikhlinin, 16 groups from Gastaldello et al. 2007 and 16 CLASH.

The results.

- **The amplitude is not decidable.** The error is 0.52, against 0.62 for one constant. The *bias*, that is, the mean offset always in the same direction, is −0.30.
- **The system-by-system prediction is not supported.** The Spearman is −0.49 over the joint sample, with different signs in the subsamples.
- **A wider edge.** 1.3·R200m describes the mean better (bias −0.04), but only *a posteriori*, that is, after seeing the data. It is an edge similar to the *splashback* radius. The splashback is the point where the matter falling into the cluster turns back for the first time: a kind of natural edge.

These outcomes hit the reading *R_b = R200m*, not the MM framework.

The MM 2.46 test remains. MM 2.46 had registered the test R2: the fixed end against measured splashback radii, on data published after 13 October 2026. This test stays unchanged.

3.6 Where the missing mass sits

The measurement. The 7 X-COP clusters with stellar profiles are looked at. The missing mass is M_missing = M_tot − ν·M_bar: the total mass minus the one given by the galaxy rule.

- The ratio between the missing mass and the stars of the member galaxies is nearly flat: the median slope, in absolute value, is 0.165 (|slope| 0.165).
- The ratio to the gas, instead, is steep: 0.625.
- In words: the missing mass is distributed like the stars of the galaxies, not like the gas.
- It holds in 6 clusters of 7. It does not hold in ZW1215.
- It also holds with a non-thermal pressure that grows with radius.

The ratio k. k = M_missing/M_star is the ratio between missing mass and stellar mass.

- At R500 it is 40. With non-thermal pressure it is 48-60.
- It is not flat: it goes from 49 to 65 and then to 40 (49 → 65 → 40), from 0.2 to 1 R500.

What it says, and what it does not.

- The test separates the shape of the stars from that of the gas.
- It does not say what the missing mass is. Other things could also follow the galaxies: halos bound to galaxies; stellar remnants from a heavier *initial mass function* (Zhang et al. 2026); cold clouds. The initial mass function says how many stars are born at each mass.
- The result contrasts with the CLASH lensing analysis of Famaey, Pizzuti and Saltas (2025). There the missing density follows the gas. But there a profile that follows the galaxies was not tried.

3.7 Galaxy groups

The data. They are 16 groups (Gastaldello et al. 2007), measured at one radius each, between 0.47 and 1.0 r500.

The results.

- At the same radius, the groups have A between 3.3 and 3.6. The X-COP clusters have 4.4-4.9.
- From 12 to 13 groups of 16 (12-13) lie below the clusters.
- The *one-sided* p-value ranges from 0.011 to 0.038. The p-value is the probability of finding such a difference by pure chance. *One-sided* means that only one direction is looked at: the groups below the clusters.
- The missing factor F is 1.8 against 1.95.
- In words: groups lack slightly less mass than clusters, but with much overlap.
- Four groups lie above the clusters.
- A group merging with another, RGH 80, has no missing mass: F = 0.83.

3.8 Readings that do not hold, or are not decidable

reading (premise)	outcome
the cluster as a full external context for its galaxies (2nd)	fails 7/7: k 4-7 against 31-88
angle of view of the long axis (5th), 16 CLASH at r2500	not decidable (Spearman +0.12 lensing, −0.13 gas; threshold 0.43)
angle at the CLASH cores	not executable (M_tot < M_bar in 10 of 13)
depth from the gas temperature, calibrated on groups	not decidable on 7 Vikhlinin clusters
depth from M500/R500	circular (total mass on both sides): not cited as a prediction
«aumenta» ("grows") (1st): the dark part per galaxy grows inside clusters	compatible only if the galaxies enter with their stars alone; not decidable with 7 clusters
epoch-equivalent a_0 (3rd, 4th): a_0 =	median cluster z ≈ 4.3 (H) or 3.2 (age), inside the registered range 1-5 (A644: 5.2 in the H form); the

$cH(z)/6$ or $a_0 \cdot t_0/t$	equivalent epoch decreases outwards, which does not show different formation epochs; found after the data, degenerate with a factor √A on the baryons

Let us read the table, row by row. The number in brackets indicates the author's premise from which the reading comes.

- **The cluster as a full external context for its galaxies** (second premise). This reading fails in 7 clusters of 7 (7/7). The ratio k comes out 4-7, while the data give 31-88.
- **The angle of view of the long axis** (fifth premise). A cluster is not a ball: it has a longer axis. A could depend on how we see it. On 16 CLASH clusters at r2500 the test is not decidable. The Spearman is +0.12 with lensing and −0.13 with the gas; the threshold was 0.43.
- **The angle at the CLASH cores.** The test cannot be run: in 10 clusters of 13 one gets M_tot < M_bar, that is, the total mass smaller than the baryonic mass.
- **The depth from the gas temperature,** with a rule calibrated on groups. On the 7 Vikhlinin clusters it is not decidable.
- **The depth from M500/R500.** It is *circular*: the total mass appears on both sides of the comparison. For this reason it is not cited as a prediction.
- **«aumenta»** ("grows") (first premise): the dark part per galaxy grows inside clusters. It is compatible only if the galaxies enter with their stars alone. With 7 clusters it is not decidable.
- **The epoch-equivalent a_0** (third and fourth premise). The idea: the cluster behaves as if it read the a_0 of a past epoch. The forms tested are $a_0 = cH(z)/6$ or $a_0 \cdot t_0/t$. Here $H(z)$ is the expansion rate at epoch z, t_0 is today's age and t the age back then.
- For the median cluster one gets $z \approx 4.3$ in the form with H, or 3.2 in the form with the age.
- It lies inside the registered range, from 1 to 5. The cluster A644 gives 5.2 in the form with H.
- The equivalent epoch decreases going outwards. This does not show different formation epochs.
- The reading was found after the data. And it is *degenerate* with a factor √A on the baryons: it gives the same result that one would get by multiplying the baryons by √A, and the data do not separate the two things.

4. Relation to MM 2.46.1

What was already published. MM 2.46 and MM 2.46.1 already published:

- the falling scale in 12 X-COP clusters;
- the law a = A/r, read as a fixed distal end;
- four registered tests for future data, called R1-R4.

Nothing here contradicts them. The data used here (X-COP, Vikhlinin 2006, Gastaldello 2007, CLASH) were published before 13 October 2026. So they do not evaluate the tests R1-R4, which stay as registered.

What is new here:

- the turned-spacetime reading of the same law;
- 19 clusters and the decomposition of the exponent;
- the cluster-by-cluster test of exponent 1;
- light against gas;
- R200m as edge;
- stars against gas;
- 16 groups;
- the readings that fail.

5. What is not claimed

- **No derivation.** The turn, $\cos \varphi = r/R_b$ and $a_0 = cH_0/6$ are readings. They were found knowing the data.
- **No distinction** from dark matter or from ordinary explanations of cluster profiles.
- **The cluster mass deficit is not solved.** What fixes R_b, and the exponent of each cluster, remain open.
- **No observational confirmation of MM.**

6. How this reading can turn out wrong

Here are the future data that can refute the reading.

- **Clusters with the hydrostatic bias calibrated with lensing.** For example the CHEX-MATE or eROSITA

samples with lensing measurements. They give an exponent for each cluster.
- If the scatter of the exponents is entirely explained by the errors, the turn with fixed R_b survives.
- If not, the turn fails in that form.
- **Measured splashback radii, against the fixed end.** It is the test R2 of MM 2.46, unchanged.
- **Groups with profiles measured at two or more radii.** The turn predicts exponent 1 in groups too, with a different R_b.
- **What the missing mass follows.** If, in samples with stellar profiles, the missing mass follows the gas instead of the galaxies, section 3.6 falls.

How the work was done

This section gathers the procedure. In the archived text it corresponds to sections 7 and 8 and to the heading; the references, at the end, are section 9.

Who did what. The framework and its premises are the author's. The author is not a professional physicist. The translations of the premises, the protocols, the code and the text were written with an AI assistant (Claude). Each step was checked by two other AI assistants, Codex ("Astra") and Gemini ("Argon"), in at most two rounds. Their reports are kept. There was no human *referee*, that is, an expert who checks a work before publication.

The registrations. The registrations are local: a SHA-256 fingerprint and the UTC time. They are not independent timestamps. A SHA-256 *fingerprint* is a code computed from the content of a file: if the file changes even a little, the fingerprint changes.

The licences. Original text and materials: CC BY 4.0. Original code: MIT licence. Third-party data are not covered by these licences and keep the conditions of their sources (`fonti/SOURCES.md`). Working language of the records: Italian.

The files cited in the sections.

- 3.1: `rapporto.py`; `note/esiti/NOTA_ESITO_RAPPORTO_v2.md`.
- 3.2: `esponente.py`; `NOTA_ESITO_ESPONENTE_v2.md`.
- 3.3: `rotazione_prova.py`, test T-R2; `NOTA_ESITO_ROTAZIONE_PROVA_v2.md`.
- 3.4: `rotazione_prova.py`, test T-R1. The A values were already in `riferimento/angolo_esterno.log` when the zones were written.
- 3.5: `rotazione.py`, `rotazione_c1.py`; `NOTA_ESITO_ROTAZIONE_v3.md`.
- 3.6: `via1_immagini.py` and corrections; `NOTA_ESITO_VIA1_v2.md`.
- 3.7: `gruppi.py`, `gruppi_c1.py`; `NOTA_ESITO_GRUPPI_v2.md`.
- 3.8, the notes by row: `NOTA_ESITO_CORONA_v3.md` (external context); `NOTA_ESITO_ANGOLO_ESTERNO_v3.md` (angle of view); `NOTA_ESITO_ANGOLO_v2.md` (angle at the cores); `NOTA_ESITO_REGOLA_A_v3.md` (depth, both rows); `NOTA_ESITO_AUMENTA_v2.md` («aumenta», "grows"); `NOTA_ESITO_TEMPO_v3.md` and `NOTA_ESITO_GALASSIE_TEMPO_v3.md` (equivalent epoch).

What is in the package.

- `codice/`: the 22 scripts, exactly as registered before each computation. The SHA-256 fingerprint and the UTC time are in `registri/REGISTRO_17_ammassi.txt`. The scripts have the paths of the author's machine.
- `riferimento/`: their *logs*, that is, the diaries that the scripts write while they work.
- `note/`: the author's premises (word for word, in `note/premesse/`), the protocols, the corrections and the outcome notes in all their versions, in Italian.
- `verifica/`: the reports of the two checking assistants and the summaries, in Italian.
- `registri/`: the registers and the SHA-256 fingerprints of the downloaded sources.
- `fonti/SOURCES.md`: sources and conditions.

How to redo the computations.

- Run `python scarica_dati.py` once. It downloads the X-COP profiles (an archive of 315 MB) and eight arXiv sources. It checks each file against `dati/MANIFEST_SHA256.txt`. The offline options are in its docstring, the description written at the start of the program.
- Then run `python riproduci.py`. It re-runs the 22 scripts in order of registration, through `esegui_uno.py`, which sets only two paths. Then it compares each log byte by byte with `riferimento/`.
- Python 3.11.4 or later, numpy, scipy and astropy are needed.

Related records. MM 2.46 (DOI 10.5281/zenodo.23340834), MM 2.46.1 (DOI 10.5281/zenodo.23506655), MM 2.44 (DOI 10.5281/zenodo.23329080), MM 2.70 (DOI 10.5281/zenodo.23730218).

This version. This student version was written with an AI assistant (Claude) from the archived text in English. First a full Italian translation was made. Then the translation was rewritten for a school student. The numbers were checked with a program: the translation against the English text, this version against the translation. The archived text is the reference. This English text is a translation of the Italian student version, which the author read and approved.

Glossary

The words of the framework (MM)

- **Distal end (R_b).** The far end of a link. In clusters it is not the cluster's outskirts. In galaxies it reaches the cosmic horizon.
- **Link.** The starting relation of MM, between two distinct terms.
- **Matrix.** Reality counted from one point, that is, as an observer counts it.
- **Premise.** A starting sentence of the author, a fixed point like a postulate.
- **Reading.** An interpretation, written so that it can be put to the test. A body *reads* something when it takes it into account.
- **Registered.** Written and fixed with its date before the computations, so that it cannot be changed afterwards.
- **Turned spacetime.** The reading of this record: the observer reads part of a cluster's spacetime as turned, with cos φ = r/R_b.

The words of physics

- **a_0.** A tiny acceleration: below a_0 gravity is stronger than Newton's. In MM $a_0 = cH_0/6$.
- **A posteriori.** Said of a check made after seeing the data.
- **Acceleration.** How much the speed changes every second.
- **Baryonic mass (M_bar).** The mass of matter made of atoms: stars and gas.
- **Boost (ν).** How many times, in the galaxy rule, gravity is stronger than Newton's.
- **χ^2 (chi-squared).** A number that measures how far a model is from the data.
- **Cosine (cos φ).** How much of an arrow turned by an angle φ remains in the starting direction: 1 for a zero angle, 0 for a right angle.
- **Dark matter.** The invisible matter that would explain the extra mass in galaxies and clusters.
- **Degrees of freedom.** The number of independent comparisons in a test.
- **Effective scale ($A \cdot a_0$).** The scale a_0 enlarged A times so that the numbers add up in a cluster.
- **Exponent.** The number that says how a quantity changes with distance. With exponent 1, A falls as 1/r.
- **Galaxy cluster.** A very large group of galaxies bound by gravity, with a very hot gas between them.
- **Galaxy group.** A set of galaxies like a cluster, but smaller.
- **Galaxy rule (RAR).** The radial acceleration relation: g_obs = g_bar/(1 − e^(−√(g_bar/a_0))).
- **Gravitational lensing.** The bending of light caused by mass. It makes it possible to weigh the invisible mass too.
- **Hydrostatic bias.** The fact that the mass obtained from the gas is usually lower than the true one.
- **Hydrostatic mass.** The mass obtained from the hot gas in balance between pressure and gravity.
- **Median.** The value that sits in the middle: half of the cases below, half above.
- **Missing factor (F).** How many times the total mass exceeds the one given by the galaxy rule.
- **MOND.** Milgrom's idea: below a_0 gravity is stronger than Newton's.
- **NFW.** A mass profile typical of dark-matter halos in computer simulations.
- **Non-thermal pressure.** The pressure due to the motions of the gas, besides that due to heat.
- **p-value.** The probability of finding such a result by pure chance.
- **R500, r2500, R200m.** Reference radii of a cluster. Within R500 the mean density is 500 times the critical density; within r2500 it is 2500 times; within R200m it is 200 times the mean density of the universe.
- **Redshift (z).** How much the light has been stretched by the expansion of the universe: the larger it is, the farther back in time the object is.
- **Spearman.** A coefficient between −1 and +1 that says how much two quantities grow together.
- **Splashback.** The radius where the matter falling into a cluster turns back for the first time.
- **Total mass (M_tot).** The mass obtained from the motion or from lensing, including the invisible part.

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